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himasree@LAPTOP-41COT5AA:~/OSLab$ nano wait.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc wait.c -o wait
himasree@LAPTOP-41COT5AA:~/OSLab$ ./wait
Child 1: PID = 569
Child 2: PID = 570
Child 3: PID = 571
Child 1 completed
Parent collected a child
Child 2 completed
Parent collected a child
Child 3 completed
Parent collected a child
All children completed.
himasree@LAPTOP-41COT5AA:~/OSLab$ nano waitpid.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc waitpid.c -o waitpid
himasree@LAPTOP-41COT5AA:~/OSLab$ ./waitpid
Child 1: PID = 582
Child 2: PID = 583
Child 3: PID = 584
Child 3 completed
Child 2 completed
Child 1 completed
Parent waited for Child 1
Parent waited for Child 2
Parent waited for Child 3
All children completed.
himasree@LAPTOP-41COT5AA:~/OSLab$ nano zombie.c
himasree@LAPTOP-41COT5AA:~/OSLab$ gcc zombie.c -o zombie
himasree@LAPTOP-41COT5AA:~/OSLab$ ./zombie
Parent: PID = 591

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Child PID = 592
Parent sleeping... Child becomes zombie.
Child: PID = 592
Child is terminating...
ps -el
Zombie removed.
himasree@LAPTOP-41COT5AA:~/OSLab$ ps -el

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F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME	CMD
4	S	0	1	0	0	80	0	-	5464	-	?	00:00:00	systemd
5	S	0	2	1	0	80	0	-	795	-	hvc0	00:00:00	init-systemd(Ub
0	S	0	6	2	0	80	0	-	795	-	hvc0	00:00:00	init
4	S	0	42	1	0	79	-1	-	12593	-	?	00:00:00	systemd-journal
4	S	0	89	1	0	80	0	-	6287	-	?	00:00:01	systemd-udev
5	S	0	98	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	99	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	100	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	102	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	104	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	106	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	108	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	109	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	110	89	0	80	0	-	6329	-	?	00:00:00	(udev-worker)
5	S	0	113	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	116	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	117	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	118	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	119	89	0	80	0	-	6251	-	?	00:00:00	(udev-worker)

5	S	0	134	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	135	89	0	80	0	-	6222	-	?	00:00:00	(udev-worker)
5	S	0	136	89	0	80	0	-	6253	-	?	00:00:00	(udev-worker)
5	S	0	137	89	0	80	0	-	6254	-	?	00:00:00	(udev-worker)
5	S	0	138	89	0	80	0	-	6254	-	?	00:00:00	(udev-worker)
5	S	0	139	89	0	80	0	-	6254	-	?	00:00:00	(udev-worker)
5	S	0	140	89	0	80	0	-	6254	-	?	00:00:00	(udev-worker)
5	S	0	141	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	142	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	143	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	144	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	145	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	146	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	147	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	148	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	149	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	150	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	152	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	153	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	154	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	155	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	156	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	157	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	158	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
5	S	0	159	89	0	80	0	-	6255	-	?	00:00:00	(udev-worker)
4	S	991	166	1	0	80	0	-	5367	-	?	00:00:00	systemd-resolve
4	S	996	170	1	0	80	0	-	22759	-	?	00:00:00	systemd-timesyn
4	S	0	179	1	0	80	0	-	1061	-	?	00:00:00	cron
4	S	101	180	1	0	80	0	-	2386	-	?	00:00:00	dbus-daemon
4	S	0	193	1	0	80	0	-	4495	-	?	00:00:00	systemd-logind